

COGNIAUDIT: Cognitive Load-Aware Auditory Feedback for Balance Control in Virtual Reality

Md. Monowar Hossain*
Dept. of Computer Science
Kennesaw State University

M. Rasel Mahmud†
Assistant Professor of Computer Science
Kennesaw State University

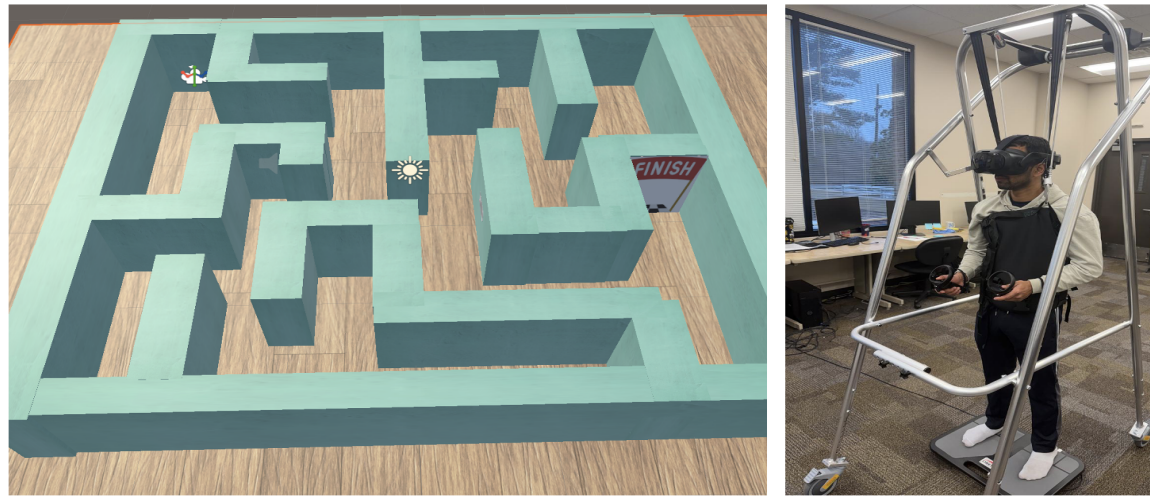


Figure 1: a) Top-down view of the virtual maze environment. b) Participants navigating the virtual maze while standing on the balance board and wearing harness for safety.

ABSTRACT

Virtual Reality (VR) systems are very useful for decision-making, training, and rehabilitation. However, their complex, multimodal settings frequently impose significant cognitive and balance difficulties. Excessive auditory, visual, and sensory stimulation can overwhelm users, impairing balance and performance and limiting the accessibility of extended reality (XR) technology. This pilot study examines the strategic design and investigation of helpful auditory feedback to mitigate cognitive load and improve balance in immersive settings. This research enhances the fundamental knowledge that auditory feedback can reduce cognitive load, with statistical analysis confirming this improvement ($p = .0011$) and improve balance in VR, providing realistic design guidelines for accessible and inclusive immersive technologies. The result also demonstrates how the body balance affects cognitive load, which helps to make XR experiences safer and more useful for everyone.

Index Terms: Cognitive Load, Virtual Reality, Auditory Feedback, Navigation

1 INTRODUCTION

Cognitive load is an important factor in how people work, feel, and make decisions. People in workplaces cope with a large amount of information, manage multiple tasks simultaneously, and make

complex decisions. These continuous stresses may cause them to feel mentally fatigued and overloaded. People may lose motivation, make more mistakes, perform less efficiently, and even have health problems when this happens.

Technologies like virtual reality (VR) create new ways to train the mind and measure how people think. These systems combine visuals, audio, and vibrotactile elements to create experiences that feel realistic, sometimes almost like real-life experiences [32]. However, when the brain receives too much information simultaneously in VR, it can lead to sensory overload. This may distract users and cause cognitive overload if the brain cannot process all the input effectively [22].

In [33], participants' body balance is degraded during the performance of a listening or spelling task involving short and long words, as well as the walking dual-task condition. The researchers found that the participant prioritized the thinking task over balance, and a higher cognitive load resulted in a decline in balance control during walking. In [16] without a feedback system in immersive VR, the standing balance generally declined, and adding assistive feedback improved the balance. The balances significantly improved when a multimodal feedback system was applied, but the cognitive load increased. They noticed that individual assistive feedback (auditory, vibrotactile, or visual) can improve balance without noticeably increasing cognitive load compared to the multimodal condition.

We investigated 15 types of auditory feedback, categorized into four types (classical, natural, electronic, and unnatural), which helped to reduce cognitive load. In the classical music conditions, four common pieces from Western classical music were included: Adagio in G minor (Albinoni), Allegro con spirito from Sonata K. 448 (Mozart), Gymnopédie No. 1 (Satie), and Für Elise (Beethoven).

*e-mail: mhossa36@students.kennesaw.edu, murad0904045@gmail.com

†e-mail: m.raselmahmud1@gmail.com, mmahmud2@kennesaw.edu

These musics were chosen because they are common for listening conditions in cognitive research [2, 4, 10, 25, 30, 35]. The natural soundscape featured a common combination of sounds, including fountains, birds tweeting, ocean waves, and birdsong. Researchers often use them to examine how they can help people feel less stressed and less mentally burdened [1, 24]. High-energy tracks such as Balearic Pumping, Natural (Imagine Dragons), Rum n' Bass (Boom Kitty), and POP/STARS (K/DA) are known as electronic music. Strong rhythmic drive, amplified bass, and fast, repetitive beats are features commonly found in this type of music that influence arousal and attentional activation. During the task, these are used to investigate how they may influence focus and reduce cognitive load [7]. There are commonly unnatural sounds like aviation noise, motorcycle noise, white noise, and construction noise used that represent the human-made acoustic environment in research to examine how they can impact stress and cognitive load [8, 16, 24]. The key contributions of our research included: 1. Investigated whether assistive sounds aid individuals in managing cognitive load in VR. 2. Differentiated between category-wise auditory feedback's impact on cognitive load. 3. Investigated the relation between balance and cognitive load.

2 BACKGROUND STUDY

2.1 Assistive Auditory Feedback Technology

When researchers work on cognitive load, they are concerned about the three types: intrinsic load (task difficulty), extraneous load (interface design), and germane load (mental processing) [34]. It is measured by assessing the subjective information (NASA Task Load Index), completion time, error rate, and physiological signals. While improving the efficiency of task completion, managing cognitive load is a vital goal of assistive auditory feedback technology. Well-designed auditory feedback cues, such as earcons, auditory icons, speech prompts, and sonification, can enhance user performance and support usability, which is related to reducing unnecessary cognitive effort [27]. One of the extended versions of sonification is the conversion of data to structured sound. Participants can easily identify the trends if the mappings are consistent and simple. The authors analyzed that if it is more complex, it increases cognitive load, which was evaluated using EEG-based workload measures [3].

Multimodal feedback can reduce task difficulty in VR tasks [9, 31]. Mahmud et al. conducted several user studies and investigated how VR-based assistive auditory, visual, vibrotactile, or multimodal feedback can improve balance and gait impairments [12, 13]. Furthermore, they [16] found that cognitive load is higher ($p = .04$) when multimodal feedback is used instead of individual feedback. Ricci et al. found that a single modality, such as auditory or haptic cues, is more effective than multimodal signals, as it helps capture attention more easily in complex situations, like navigation. Auditory feedback can enhance learning and memory by reinforcing task-relevant cues [11] and improve standing balance [6, 17]. Real-world studies demonstrate that auditory assistive cues can improve navigation, attention management, and error correction [18, 23].

Although complex sound can affect the cognitive load for the navigation system, for those who struggle with visualization, auditory feedback helps navigate by detecting obstacles and plotting a path [28]. For a safety-critical environment like automated driving, the "speech" and "spearcon" cues performed better than earcons, as they helped to understand the critical situation faster and resolve it with less mental stress [5].

2.1.1 Classical Music as an Auditory Feedback

Mozart's Sonata is the most popular, which is used for research in the sectors of music, cognition, and the brain. The authors reviewed the past 30 years of studies on humans and animals that investigated how the person's excitement, preferences, and listening context affected their memory, attention, and brain activity [2]. "Mozart's Sonata and

Adagio in G Minor" music was used in the study, and the researcher examined how these influenced the working memory in different age groups. They also found that there is no overall effect on complex tasks [4]. In [35], for improving learning and task performance, the music of "Mozart's Sonata" is a better solution than other reversed versions that have increased performance. The authors investigated that Mozart's Sonata provides a more positive effect on brain activity than others. In [10], while solving spatial rotation tasks and math tasks, the brain activity is affected by listening to Mozart's music. This music degraded the result of the math task, but it provided a better performance in the spatial task. The researchers found that music can boost activity in the brain area that assists in combining information into meaningful patterns. Shen et al. [30] also investigated how the soft background music (Gymnopedie No.1) without lyrics affects participants' attention and showed that this music performs better than silence. They suggested that soft background music can improve participants' attention and quick learning. Mohammadian et al. [25] explored how listening to "Für Elise" music in noisy simulated open-plan offices contributed to improving the mental workload more than using irrelevant speech. They used n-back tests to assess performance accuracy and provided suggestions that the background music can act as a countermeasure to noise and provide additional support for cognitive performance. We preferred auditory over vibrotactile [19–21] or visual feedback [14, 15], as vibrotactile feedback requires participants to wear vibrating devices and visual feedback requires them to look at visual signs, which may increase cognitive load.

2.1.2 Natural Music as an Auditory Feedback

The author in [1] analyzed that a mixture of sounds from a fountain and tweeting birds can reduce stress levels compared to noisy sounds. Moreover, nature sounds can help recover skin conductance levels faster after completing a stressful task. They also suggested that for participants' relaxation, natural sounds will play a vital role. After stressing and during rest, how different natural sounds, such as "ocean waves and birds' songs", affected the participants' bodies, as analyzed in [24]. They also investigated the fact that heart rate degrades after completing a stressful task, and skin conductance improves when participants hear natural sounds during rest, which helps reduce the cognitive load level.

2.1.3 Electronic Music as an Auditory Feedback

When the participants performed the VR game Beat Saber, the authors investigated how musical training affected their cognitive load and task performance. In the musical training, they used the most common music: "Balearic Pumping, Natural, Rum n' Bass, and POP/STARS". Moreover, they observed that the harder level of the task required more mental effort, but musical training could help reduce the cognitive load [7].

2.1.4 Unnatural Music as an Auditory Feedback

In the Medvedev et al. [24] study, aviation, motorcycle, and construction sounds are considered unnatural sounds that negatively affect the participants' perceptual and physiological strain. Participants generally considered these sounds calming and restorative, comparing them to natural sounds, such as birds' songs and ocean waves. The findings of this study were that these sounds have an impact on greater physiological strain (higher skin conductance) and a slow rate of stress recovery. Moreover, they suggested that it may elevate the cognitive load. The researchers [8] sought to determine whether the addition of white noise could impact children's thinking and attention at varying levels of ability. They suggested, based on the Moderate Brain Arousal model, that every person has a different amount of natural neural noise and that improving children's brain activity can be achieved by adding external white noise. Mahmud et al. [16] also included white noise in their study to enhance

participant stability and comfort.

3 METHODS

3.1 Hypotheses

Based on the literature we had the following hypothesis:

- H1: Without assistive auditory feedback in VE, the cognitive load will increase. When a participant performs a task without feedback, it increases the cognitive load [26] because it takes more time to make a decision that is best suited to that situation.
- H2: We hypothesize that the participant's postural balance is related to the brain activity when he performs a task. Balance improves with reduced cognitive load, and vice versa. Small et al. [33] analyzed that when a participant performs a cognitive task while walking, their ability to maintain dynamic balance decreases under a higher cognitive load.
- H3: Participants exposed to classical and natural category audio will experience significantly lower cognitive load than when exposed to electronic and unnatural category audio.
- H4: There will be a significant difference in cognitive load across the different audio categories (classical, natural, electronic, and unnatural).

3.2 System Description

3.2.1 Balance Measurement

We measured the participant's balance using the BTrackS Balance Plate, which operates at a 25 Hz sampling frequency.

3.2.2 Safety Equipment

We utilized a harness system to ensure participants' safety. When a participant conducts the VR task, there is a risk of falling, so we used this system to mitigate that risk.

3.2.3 Computers, VR Equipment, and Software

We used the HTC VIVE Focus Vision to view the virtual maze. It features noise-canceling headphones to mitigate external audio noise and ensure the realism of virtual environments (VE). We collected BTrackS Balance Plate data using NI LabVIEW (2020), which was stored on a high-performance computer equipped with an Apple M3 chip, an 8-core CPU, a 10-core GPU, and 8 GB of RAM. We synchronize VR runtime data with balance data through timestamp alignment.

3.2.4 Environment

We performed our task in a lab that was over 1,000 square feet in size. The experiment was conducted in a calm, temperature-controlled space with no external distractions.

3.2.5 Participants

We recruited six participants, all of whom were adults aged 18 and older, from diverse backgrounds. The detailed information for the participants is shown in Table 1.

Table 1: Descriptive statistics for participants

Participants		Age (years)		Height (cm)		Weight (lb)	
Male	Female	Mean	SD	Mean	SD	Mean	SD
4	2	27	2.61	169.67	9.37	183	27.71

3.3 Experimental Conditions

3.3.1 Auditory Feedback in VR

We investigated the impact of 15 types of auditory feedback (classified as classical, natural, electronic, and unnatural) on cognitive load during VR-based navigation tasks that improved balance.

3.3.2 No Feedback in VR

The participant played the task in VE without the involvement of any additional audio feedback, considered as a baseline condition.

3.4 Metrics

3.4.1 Cognitive Load Assessment

After completing each task, participants were asked to fill out the NASA Task Load Index (NASA-TLX) form on a scale of 0 to 10. From this self-reported evaluation form, we could gauge the cognitive load with each condition.

3.4.2 Center of Pressure (CoP) Velocity

The term CoP velocity refers to the information about the dynamic shifts of the body's center of pressure during the participant's involvement in various tasks or conditions. For each condition, this metric was systematically captured and utilized to monitor and measure changes in balance.

3.5 Study Procedure

3.5.1 Experimental Setup

The participant wore a headset and used the safety harness to ensure stability and safety throughout the sessions. For monitoring the participant's balance, a balance board was used during all conditions.

3.5.2 Questionnaires

The NASA-TLX form was completed by the participants after finishing each session.

3.5.3 Experimental Tasks

Participants performed the maze navigation task in VR. To control for potential order effects, the auditory feedback was provided in a counterbalanced order.

3.5.4 Baseline Measurements

The participant performed the maze navigation task without any auditory feedback. Each trial lasts about one minute.

3.5.5 Tasks in Virtual Reality

The VR task is similar to the baseline activity, including the additional auditory feedback conditions.

3.5.6 Maze Navigation in VR

The VR Maze Game was chosen as the experimental setting because navigating a maze requires various cognitive functions, such as working memory, attention control, and decision-making, all of which are important to cognitive load. To navigate a maze, participants must constantly encode and remember spatial information, plan their paths, track their mistakes, and adjust to varying audio cues [29]. Using HTC Vive controllers, the participant moved and turned in the maze environment while standing on the balance board. The left controller was used for moving, and the right controller was used for turning in the path. In auditory feedback, we provided 15 different types of music in the left and right ears, which indicated to turn left and right, respectively, while the auditory feedback originating from the front indicated to move forward.

4 RESULTS

4.1 Effect of Auditory Feedback on Cognitive Load

The repeated-measures ANOVA provided a significant difference in the impact of different auditory feedback on cognitive load, $F(4, 20) = 7.02, p = .0011$. From the paired t-tests result, we got significant p -values of 0.046, 0.001, and 0.004 for classical vs electronic, classical vs unnatural, and natural vs unnatural, respectively, and other results are marginally significant. The result also demonstrated that classical and natural audio feedback conditions reduced the cognitive load

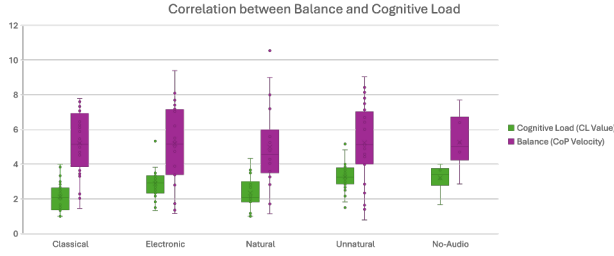


Figure 2: Balance and Cognitive Load for Different Types of Auditory Feedback

more than the no-audio feedback condition. Classical Music (Mean, $M = 2.12$) and Natural Music ($M = 2.34$) have lower cognitive value than Electronic Music ($M = 2.89$), and the highest is for Unnatural Music ($M = 3.27$). Thus, the result supported our hypothesis H4.

4.2 Pairwise Comparisons Against No-Audio Baseline

We compared in Table 2 each sound condition with the no-audio baseline using a t-test. This analysis demonstrated that classical music ($t=-2.47$, $p=.057$, $d_z=-1.01$) and natural music ($t=-2.33$, $p=.068$, $d_z=-0.95$) presented marginally significant improvements in cognitive load. On the other side, Electronic Music ($t=-1.40$, $p=.221$, $d_z=-0.57$) and Unnatural Music ($t=0.16$, $p=.875$, $d_z=0.07$) had no meaningful difference compared to the baseline condition. This analysis established that classical and natural audio categories have a greater effect on cognitive load than other categories, thus supporting H3.

Table 2: Correlation between balance and cognitive load for various auditory conditions

Comparison	t_{Bal}	p_{Bal}	d_z^{Bal}	t_{CgLd}	p_{CgLd}	d_z^{CgLd}	$r^{(Bal,CgLd)}$	p_r
Classical vs No-Audio	-0.11	0.918	-0.04	-2.47	0.0568	-1.01	-0.82	0.0480*
Electronic vs No-Audio	-0.17	0.874	-0.07	-1.40	0.2209	-0.57	-0.47	0.3507
Natural vs No-Audio	-0.46	0.666	-0.19	-2.33	0.0676	-0.95	-0.98	0.00048*
Unnatural vs No-Audio	-0.06	0.954	-0.02	0.16	0.875	0.07	-0.72	0.1072

Note. t = paired-sample t statistic; p = probability value; d_z = Cohen's effect size for paired samples; r = Pearson correlation between balance and cognitive load.

4.3 Relationship Between Cognitive Load and Balance

The correlation analysis in Table 2 pointed out that cognitive load and postural balance have a reverse relationship. Classical Music ($r=-0.82$, $p=.048$) and Natural Music ($r=-0.98$, $p<.001$) have statistically significant results, while Electronic Music ($r=-0.47$, $p=.351$) and Unnatural Music ($r=-0.72$, $p=.107$) showed weaker and non-significant trends. Therefore, it confirmed that a higher cognitive load is associated with lower postural balance, supporting H2.

4.4 Assistive Auditory Feedback vs No-Auditory Feedback

When compared to the No-Audio feedback, assistive auditory navigation feedback, especially classical and natural music significantly reduced cognitive load in Figure 2. Since cognitive load was strongly related to balance, these reductions indirectly helped maintain postural stability. So, auditory feedback made VR navigation tasks easier to think about, which supports H1.

5 DISCUSSION

The preliminary results indicated that classical and natural audio categories produced a lower cognitive load score relative to no-audio feedback and other audio categories. Moreover, the results also revealed the negative correlation between cognitive load and postural balance (balance decreases with increased cognitive load), which will motivate us to run a user study with more participants to further investigate the interesting findings.

6 CONCLUSION

This pilot study highlighted that the assistive auditory feedback is an effective approach for improving cognitive load, which is strongly associated with improved balance. Classical and natural audio category was more effective than other auditory feedback for the cognitive load. These preliminary results established that the selection of auditory cues is an important factor in designing VR systems. The addition of naturalistic audio feedback in VR environments helps participants to become more efficient, safer, and more accessible, especially those who are vulnerable to cognitive overload or balance impairment.

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